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ORIGINAL RESEARCH ARTICLE

Risk of Heart Failure Hospitalization for GLP-1 Receptor Agonists Versus DPP-4 Inhibitors or SGLT-2 Inhibitors in Patients With Type 2 Diabetes: A Target Trial Emulation

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BACKGROUND: Novel treatments are needed for the primary and secondary prevention of heart failure in patients with type 2 diabetes, including individuals with and those without a history of heart failure. Conflicting trial evidence exists on whether glucagon-like peptide-1 receptor agonists (GLP-1RAs) reduce the risk of hospitalization for heart failure (HHF) in this broad population and whether this is a class effect or varies by specific agent. Furthermore, their comparative effectiveness against sodium-glucose cotransporter-2 inhibitors (SGLT-2is) is unknown.

METHODS: We emulated 2 target trials using population-based health care data from Stockholm, Sweden (2010–2021). Target trial 1 included adult patients with type 2 diabetes who newly initiated GLP-1RA versus dipeptidyl peptidase-4 inhibitors (DPP-4is), and target trial 2 compared GLP-1RA with SGLT-2i. The primary outcome was HHF. Cox regression was used to estimate intention-to-treat hazard ratios, with inverse probability of treatment weighting used to balance 72 confounders. Major adverse cardiovascular events were used as the positive control outcome. Analyses were conducted for the GLP-1RA class overall and for individual agents, including liraglutide and semaglutide.

RESULTS: Target trial 1 included 32 979 patients (42% GLP-1RA and 58% DPP-4i) with a mean age of 64 years and 40% female; and target trial 2 included 30 104 patients (49% GLP-1RA and 51% SGLT-2i) with a mean age of 63 years and 38% female. Starting a GLP-1RA was associated with a lower 3-year absolute risk of HHF than starting a DPP-4i (3.4% versus 4.3%), corresponding to a weighted hazard ratio of 0.77 (95% CI, 0.66–0.91). Absolute 3-year risks for GLP-1RA compared with SGLT-2i on HHF were 3.6% and 3.3% with a weighted hazard ratio of 1.02 (95% CI, 0.85–1.18). The absolute risk difference was largest for patients with higher predicted risk of heart failure at baseline. Results were consistent for single agents, in per-protocol analyses, and in the majority of subgroups. In positive control outcome analyses, GLP-1RA use was associated with a lower rate of major adverse cardiovascular events than DPP-4i use (weighted hazard ratio, 0.85 [95% CI, 0.74–0.99]), consistent with trial findings.

CONCLUSIONS: GLP-1RA use is associated with a lower risk of HHF compared with DPP-4i and similar risks compared with SGLT-2i in routine clinical care for patients with type 2 diabetes.

Key Words: cardiovascular diseases ■ clinical trial ■ hypoglycemic agents ■ metabolic diseases

Editorial, see p 1101

Type 2 diabetes (T2D) is expected to affect >592 million people worldwide by 2035, thus rendering T2D one of the most important public health challenges over the next decade.[1] Patients with diabetes have more than twice the risk of developing heart failure compared with those without diabetes.[2,3] Therefore, there is an urgent

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Clinical Perspective

What Is New?

- In this real-world study including >30 000 patients with type 2 diabetes, use of glucagon-like peptide-1 receptor agonists was associated with a significantly lower risk of heart failure hospitalization compared with use of dipeptidyl peptidase-4 inhibitors, with the largest absolute risk difference observed for patients who had the largest predicted baseline risk of heart failure.
- Overall, glucagon-like peptide-1 receptor agonists were associated with similar risks of heart failure hospitalization compared with sodium-glucose cotransporter-2 inhibitors, with consistent findings for liraglutide and semaglutide.

What Are the Clinical Implications?

- Glucagon-like peptide-1 receptor agonists are associated with lower risks of both heart failure and atherosclerotic events in broad populations with type 2 diabetes from routine clinical practice.
- These real-world findings suggest that glucagon-like peptide-1 receptor agonists may be considered as an alternative therapeutic option to sodium-glucose cotransporter-2 inhibitors to reduce the risk of heart failure hospitalization in patients with type 2 diabetes, both with and without established heart failure, which needs to be confirmed in randomized controlled trials.

Nonstandard Abbreviations and Acronyms

BMI	body mass index
DPP-4i	dipeptidyl peptidase-4 inhibitor
eGFR	estimated glomerular filtration rate
FLOW	Evaluate Renal Function With Semaglutide Once Weekly
FREEDOM-CVO	Future Revascularization Evaluation in Patients With Diabetes Mellitus: Optimal Management of Multivessel Disease-Cardiovascular Outcome
GLP-1RA	glucagon-like peptide-1 receptor agonist
HbA1c	glycated hemoglobin
HHF	hospitalization for heart failure
HR	hazard ratio
LEADER	Liraglutide Effect and Action in Diabetes: Evaluation of Cardiovascular Outcome Results
MACE	major adverse cardiovascular event
PIONEER 6	Peptide Innovation for Early Diabetes Treatment 6
PP	per-protocol
PREVENT-HF	Predicting Risk of Cardiovascular Disease EVENTs-Heart Failure
RD	risk difference
SCREAM	Stockholm Creatinine Measurements
SGLT-2i	sodium-glucose cotransporter-2 inhibitor
T2D	type 2 diabetes
TRS-HF$_{DM}$	Thrombolysis in Myocardial Infarction Risk Score for Heart Failure in Diabetes
UACR	urinary albumin-to-creatinine ratio

need for novel treatment approaches for the primary and secondary prevention of heart failure in patients with T2D.

Glucagon-like peptide-1 receptor agonists (GLP-1RAs) have been shown to reduce the risk of major adverse cardiovascular events (MACEs), all-cause mortality, and kidney outcomes in patients with T2D.[4–7] However, there is conflicting evidence on the effects of GLP-1RA on heart failure outcomes for patients with T2D. A meta-analysis of 8 large cardiovascular outcome trials showed that GLP-1RAs were associated with an 11% lower hazard of hospitalization for heart failure (HHF) in patients with T2D.[6] A more recent meta-analysis that included data from the FLOW trial (Evaluate Renal Function With Semaglutide Once Weekly) and FREEDOM-CVO trial (Future Revascularization Evaluation in Patients With Diabetes Mellitus: Optimal Management of Multivessel Disease-Cardiovascular Outcomes) reported a 13% lower hazard.[8] In addition, more recent data have shown that GLP-1RAs improve heart failure outcomes in specific populations such as those with obesity,[9] diabetic kidney disease,[10] or heart failure with preserved ejection fraction.[11] However, another meta-analysis that also included smaller trials showed no effects of GLP-1RAs on HHF in patients with T2D.[12]

In the presence of conflicting clinical trial evidence, observational studies may help to determine the effectiveness of GLP-1RAs in broad, unselected populations that are reflective of routine clinical practice. Furthermore, observational studies allow the comparison of GLP-1RA against other guideline-recommended treatments rather than placebo. We therefore investigated the comparative effectiveness of GLP-1RAs and dipeptidyl-peptidase 4 inhibitors (DPP-4is) or sodium-glucose cotransporter-2 inhibitors (SGLT-2is) on the risk of HHF among adults with T2D in a real-world setting.

METHODS

Data Source

We used data from the SCREAM project (Stockholm Creatinine Measurements), a healthcare use cohort including all adult

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residents in Stockholm between 2006 and 2021.[13] The region of Stockholm had a population of 2.3 million citizens in 2021 and provides universal health care with a single unified health system. Administrative databases with complete information on demographic data, health care use, diagnoses, therapeutic/surgical procedures, and vital status were enriched with performed laboratory tests, dispensed prescriptions at Swedish pharmacies, and validated kidney replacement therapy end points. Registries were linked and deidentified by the Swedish National Board of Welfare. Because the study used deidentified data, it was deemed not to require informed consent and was approved by the regional ethical review boards and the Swedish National Board of Welfare (2017/793-31). The study follows the Strengthening the Reporting of Observational Studies in Epidemiology reporting guidelines.[14] The data are not publicly available but may be shared on reasonable request (J.-J.C., juan.jesus.carrero@ki.se).

Target Trial Specification and Emulation

Following the target trial emulation framework,[15] we specified the protocols of 2 separate target trials that would evaluate the comparative effectiveness of GLP-1RA versus DPP-4i (target trial 1) and GLP-1RA versus SGLT-2i (target trial 2) on the risk of HHF in patients with T2D. Target trials 1 and 2 address distinct research questions. In target trial 1, we used DPP-4i (excluding saxagliptin) as a neutral comparator to estimate the effect of GLP-1RA versus placebo-like conditions because randomized trials and large-scale observational safety studies have shown that DPP-4i does not affect heart failure risk.[16–18] In target trial 2, we selected SGLT-2i, a guideline-recommended therapy with proven benefits on heart failure,[19] to assess the effect of GLP-1RA versus established standard of care. The key components of the target trial protocols and their emulation process are described later and in Table 1. We chose not to emulate a single 3-arm trial of GLP-1RA versus DPP-4i versus SGLT-2i because this would require patients to not have used any of the 3 drugs. Such a target trial would reduce generalizability and statistical power because many patients in clinical practice already use a DPP-4i and subsequently switch to or add a GLP-1RA or SGLT-2i.

Eligibility Criteria

Patients who fulfilled all eligibility criteria between January 1, 2010, and December 31, 2021 (target trial 1), or between January 1, 2014, and December 31, 2021 (target trial 2), were included in this study. Dates for inclusion differed between the target trials because SGLT-2is were first approved in Europe in 2013. Inclusion criteria were age ≥18 years with a diagnosis code for T2D. To ensure that we would include new users of the medications under comparison, eligible patients were required to have no filled prescription in the previous year of GLP-1RA and DPP-4i (for target trial 1) or GLP-1RA and SGLT-2i (for target trial 2). For these 2 target trials, we excluded patients who had diagnosis codes or laboratory values indicating kidney failure (ie, dialysis, kidney transplantation, or estimated glomerular filtration rate [eGFR] <15 mL·min^{-1}·1.73 m^{-2}), pancreatitis, cirrhosis, acute hepatitis, multiple endocrine neoplasia type 2, or other organ transplantations (definitions provided in Table S1 and Figure S1). We also excluded individuals who had <1 day of follow-up.

Treatment Strategies

We chose an active-comparator new user design for both trials because it provides relevant information on the comparative effectiveness of drugs and reduces confounding by indication compared with nonactive comparator designs.[20] The treatment strategies of interest for target trial 1 were "start a GLP-1RA and continue use during follow-up" versus "start a DPP-4i and continue use during follow-up." For GLP-1RA, we included patients initiating exenatide, liraglutide, lixisenatide, dulaglutide, or semaglutide, whereas for DPP-4i, we included sitagliptin, vildagliptin, or linagliptin. We excluded saxagliptin because of previous potential safety concerns of increased heart failure risk for saxagliptin.[21,22] For the other DPP-4is, there were no safety concerns in randomized controlled trials and large-scale observational studies.[16–18,23,24] Similarly, for target trial 2, treatment strategies of interest were "start a GLP-1RA and continue use during follow-up" versus "start an SGLT-2i and continue use during follow-up." For SGLT-2i, we included patients initiating dapagliflozin, canagliflozin, empagliflozin, and ertugliflozin. Other specific GLP-1RAs (eg, albiglutide), SGLT-2is (eg, sotagliflozin), and DPP-4is (eg, alogliptin) were not used in Sweden during the time period.

Treatment Assignment

Patients who newly filled a prescription for GLP-1RA or DPP-4i during the study period were assigned to target trial 1, and patients newly initiating GLP-1RA or SGLT-2i were assigned to target trial 2.

Outcomes and Study Covariates

The primary outcome was HHF, defined as hospitalization with a diagnosis code for heart failure in the first or second position (definitions in Table S2).

We extracted a number of covariates prior to the index date (definitions provided in Table S3). These included age, sex, duration of diabetes, calendar year, marital status, living situation, education level, household disposable income, laboratory measurements (eg, glycated hemoglobin [HbA1c], eGFR, and urinary albumin-to-creatinine ratio [UACR]), comorbidities, use of diabetes drugs (eg, metformin, sulfonylureas, insulin), use of cardiovascular drugs (eg, angiotensin-converting enzyme inhibitors/angiotensin receptor blockers, β-blockers, mineralocorticoid receptor antagonists), and health care use (eg, number of hospitalizations and number of outpatient visits in the previous year). eGFR was calculated from creatinine with the Chronic Kidney Disease Epidemiology Collaboration 2021 equation.[25,26] When UACR was not available, it was estimated from urinary protein-to-creatinine ratio or dipstick using previously published conversion formulas by Sumida et al,[27] which include sex, hypertension, and diabetes.

Start and End of Follow-Up

The date of the first GLP-1RA, DPP-4i, or SGLT-2i dispensation was defined as the start of follow-up (ie, the index date, time 0). Patients were followed up until the occurrence of an outcome, death, emigration out of Stockholm region, administrative end of the study period (December 31, 2021), or up to 3 years of follow-up, whichever occurred first.

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Table 1. Target Trial Specification and Its Emulation Using Observational Data From the SCREAM Project

Target trial component		Specification	Emulation with SCREAM
Causal estimand	Eligibility criteria	During the study period between January 1, 2010, and December 31, 2021, for GLP-1RA vs DPP-4i (target trial 1) or between January 1, 2014, and December 31, 2021, for GLP-1RA vs SGLT-2i (target trial 2) Inclusion criteria: 1. Age ≥18 y 2. Diagnosis of T2D 3. No prior use of GLP-1RA and DPP-4i (target trial 1) or GLP-1RA and SGLT-2i (target trial 2) in the past year	Same as target trial. Each eligibility criterion is explicitly mapped to corresponding observational data items in Table S1.
		Exclusion criteria: 1. History of kidney failure (initiation of dialysis or kidney transplantation or eGFR <15 mL·min^{-1}·1.73 m^{-2}) 2. History of pancreatitis 3. History of cirrhosis 4. History of acute hepatitis 5. History of multiple endocrine neoplasia type 2 6. History of organ transplantation 7. <1 d of follow-up	
	Treatment strategies	Target trial 1: 1. Start GLP-1RA* and continue use during follow-up 2. Start DPP-4i* and continue use during follow-up Target trial 2: 1. Start GLP-1RA* and continue use during follow-up 2. Start SGLT-2i* and continue use during follow-up	Same as target trial
	Assignment (unmasked)	Randomization. Patients are aware to which strategy they are assigned (ie, no blinding).	Patients are assigned to the treatment strategy that their observational data are compatible with at baseline (eg, patients who filled a prescription for GLP-1RA were assigned to the GLP-1RA strategy).
	Outcomes	HHF	Same as target trial. The study outcome is operationalized for observational data as detailed in Table S2.
	Start/end of follow-up	Follow-up is started at treatment assignment and ends on the occurrence of an outcome event, death, emigration from Stockholm region, administrative end of data collection (December 31, 2021), or up to 3 y of follow-up, whichever occurs first.	Same as target trial
	Causal contrast of interest	Intention-to-treat effect and PP effect	Observational analog of the intention-to-treat and PP effect
Identifying assumptions	Intention-to-treat effect	Baseline randomization	Conditional exchangeability assumption for treatment†
	PP effect	Baseline randomization and sequential conditional exchangeability assumption for censoring due to treatment discontinuation or switching‡	Conditional exchangeability assumption for treatment† and sequential conditional exchangeability assumption for censoring due to treatment discontinuation or switching‡
Estimator	Intention-to-treat analyses	Patients are analyzed according to their original treatment assignment regardless of treatment discontinuation or switching. Intention-to-treat effects, measured as HR and RDs at 6-mo intervals, are estimated by Cox proportional hazard regression model and Aalen-Johansen estimator, respectively.	Baseline randomization is emulated by adjusting for all baseline confounders in Table S3 through inverse probability of treatment weighting. Intention-to-treat effects, measured as HR and RDs at 6-mo intervals, are estimated by weighted Cox proportional hazard regression model and weighted Aalen-Johansen estimator, respectively.§
	PP analyses	Patients are censored when they discontinue their originally assigned treatment or switch to the comparator treatment. Inverse probability of censoring weighting is used to adjust for the selection bias introduced by the censoring. PP effects, measured as HR and RDs at 6-mo intervals, are estimated with weighted pooled logistic regression model and product-limit estimator, with detailed methods provided in Supplemental Methods.‡	Same as the target trial
	Analyses of single GLP-1RA agents	Liraglutide vs DPP-4i Liraglutide vs SGLT-2i Semaglutide vs DPP-4i Semaglutide vs SGLT-2i	Same as the target trial

(*Continued*)

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Table 1. Continued

Target trial component		Specification	Emulation with SCREAM
	Subgroup analyses	Age (≥70 y vs <70 y) Sex (male vs female) Calendar year (2010–2017 vs 2018–2021 for target trial 1 or 2014–2017 vs 2018–2021 for target trial 2) Baseline eGFR (≥60 mL·min^{-1}·1.73 m^{-2} vs <60 mL·min^{-1}·1.73 m^{-2}) ASCVD at baseline (yes vs no) Heart failure at baseline (yes vs no) SGLT-2i use at baseline (yes vs no for target trial 1) DPP-4i use at baseline (yes vs no for target trial 2) ACEi/ARB use at baseline (yes vs no) β-Blocker use at baseline (yes vs no) MRA use at baseline (yes vs no)	Same as the target trial
	Sensitivity analyses (except PP analyses)	None	To test the robustness of study results influenced by 1. Outcome definition: A more stringent HHF definition as detailed in Table S2 2. Follow-up period: A minimum follow-up of 1 y by restricting to patients with index dates before January 1, 2021 3. Unobserved confounding (violation of conditional exchangeability assumption for treatment*): Positive control outcome: MACEs Quantitative bias analyses for unobserved BMI 4. Heterogeneity of treatment effect according to baseline heart failure risk among heart failure–naive patients: TRS-HF$_{DM}$ 5. eGFR equation: LM EKFC
	Missing data	Multiple imputation by classification and regression trees with 5 imputations	Same as target trial

ACEi indicates angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; ASCVD, atherosclerotic cardiovascular disease; BMI, body mass index; DPP-4i, dipeptidyl-peptidase 4 inhibitor; eGFR, estimated glomerular filtration rate; EKFC, European Kidney Function Consortium; GLP-1RA, glucagon-like peptide-1 receptor agonist; HHF, hospitalization for heart failure; HR, hazard ratio; LM, revised Lund-Malmo; MACE, major adverse cardiovascular event; MRA, mineralocorticoid receptor antagonist; PP, per-protocol; RD, risk difference; SCREAM, Stockholm Creatinine Measurements; SGLT-2i, sodium-glucose cotransporter-2 inhibitor; T2D, type 2 diabetes; and TRS-HF$_{DM}$, Thrombolysis in Myocardial Infarction Risk Score for Heart Failure in Diabetes.

*GLP-1RAs included exenatide, liraglutide, lixisenatide, dulaglutide, and semaglutide. DPP-4is included sitagliptin, vildagliptin, and linagliptin, with saxagliptin excluded because of prior safety concerns regarding heart failure. SGLT-2is included dapagliflozin, canagliflozin, empagliflozin, and ertugliflozin. Other specific agents within these classes (eg, albiglutide, sotagliflozin, alogliptin) were not available in Sweden during the study period.

†The assumption of conditional exchangeability for treatment requires that patients in different treatment strategy groups are exchangeable at time 0 conditional on all baseline covariates listed in Table S3;

‡The assumption of sequential conditional exchangeability for censoring due to treatment discontinuation or switching requires that, at each moment during follow-up, censored patients are exchangeable with uncensored patients conditional on treatment, time-fixed covariates, and time-varying covariates listed in Table S3.

§In all weighted analyses, 95% CIs were estimated with nonparametric bootstrapping, constructing 500 bootstrap samples from each imputed dataset to generate a total of 2500 bootstrap replicates.

Causal Contrast

Our main analysis used an intention-to-treat approach, which analyzes patients according to their original treatment assignment at baseline, regardless of treatment discontinuation or switching during follow-up. Intention-to-treat analyses can bias findings toward the null when a large proportion of patients discontinue their drug during follow-up, which is common in routine clinical practice. We therefore also performed a per-protocol (PP) analysis that estimates the effect of following the treatment strategy during follow-up, that is, continued GLP-1RA use.[28]

Statistical Analyses

We used inverse probability of treatment weighting to adjust for 72 measured baseline confounders. Using a multivariable logistic regression model, we estimated the probability of initiation of GLP-1RA as a function of the baseline covariates listed in Table S3. Weights were stabilized by adding the marginal probability of the received treatment to the numerator of the weights. We calculated standardized mean differences to assess the balance of covariates between the treatment groups before and after weighting, using a standardized mean difference of >0.1 to indicate meaningful imbalance. We estimated weighted incidence rates per 1000 person-years based on a Poisson distribution and weighted hazard ratios (HRs) from weighted Cox proportional hazard regression models. We used the weighted Aalen-Johansen estimator to estimate absolute risks for each treatment arm during follow-up, which appropriately accounts for the competing risk of death. From these absolute risks, we also calculated risk differences (RDs) at 6-month intervals. Detailed information about the calculation process of the Aalen-Johansen estimator is included in the Supplemental Methods.

There were no missing data for covariates other than socioeconomic status and laboratory measurements. Marital status,

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living situation, education level, household disposable income, HbA1c, eGFR, and UACR were missing in 10.1%, 1.1%, 3.5%, 1.1%, 11.2%, 12.3%, and 31.5% of patients in target trial 1 and in 7.5%, 1.2%, 3.3%, 1.2%, 10.3%, 11.3%, and 26.5% in target trial 2, respectively. We therefore used multiple imputation by chained equations with classification and regression trees to impute 5 complete datasets separately for each target trial. The imputation model included the treatment variable, all covariates, the event indicator for the outcome, and the Nelson-Aalen estimate of the baseline cumulative hazard. Additional information on the order of multiple imputation, bootstrapping,[29] and model fitting is given in the Supplemental Methods.

Analyses of Single GLP-1RA Agents, Effect Modification by Baseline Heart Failure Risk, and Subgroup Analyses

To investigate potential differences between GLP-1RA, we also assessed the effects of liraglutide and semaglutide separately (the most commonly used GLP-1RA) by constructing 4 new cohorts: liraglutide versus DPP-4i, liraglutide versus SGLT-2i, semaglutide versus DPP-4i, and semaglutide versus SGLT-2i. All statistical analyses (propensity score estimation, multiple imputation, bootstrapping, etc) were repeated in these new cohorts. For the analyses of semaglutide, we included patients from January 1, 2018, because semaglutide was first approved in Europe in 2017 and truncated follow-up at 2 years (versus 3 years in main analysis) because few patients had follow-up beyond 2 years.

To investigate the utility of a risk-based approach for primary heart failure prevention,[30] we restricted our analysis to patients without a history of heart failure and stratified them according to baseline heart failure risk as quantified by the Thrombolysis in Myocardial Infarction Risk Score for Heart Failure in Diabetes (TRS-HF_{DM}).[31] Within each risk stratum (0 [low], 1 [intermediate], and ≥2 [high] points), we repeated the main analysis (HHF) and performed an analysis of the positive control outcome (MACEs) to evaluate whether there was effect modification across baseline risk of heart failure on these outcomes. Furthermore, we performed a number of subgroup analyses in the following prespecified strata: age (≥70 years versus <70 years), sex, calendar year (2010–2017 versus 2018–2021 for target trial 1 or 2014–2017 versus 2018–2021 for target trial 2), baseline eGFR (≥60 mL·min^{-1}·1.73 m^{-2} versus <60 mL·min^{-1}·1.73 m^{-2}), presence or absence of atherosclerotic cardiovascular disease at baseline, heart failure, use of an SGLT-2i (for target trial 1) or a DPP-4i (for target trial 2), angiotensin-converting enzyme inhibitor/angiotensin receptor blockers, β-blocker, and mineralocorticoid receptor antagonists.

Sensitivity Analyses

We performed the following sensitivity analyses to test the robustness of our study results. First, we performed PP analyses to estimate the effect of following the treatment strategy during follow-up, which accounts for treatment discontinuation or switching. Briefly, patients were censored when they discontinued their initially assigned drug or started the comparator drug during follow-up. Discontinuation of treatment was defined as absence of a new dispensation within 90 days after the estimated supply of the most recent dispensation had ended. Because we expected the rate of treatment discontinuation or switching to be dependent on the initial treatment assigned and patient characteristics, we used inverse probability of censoring weighting to account for the selection bias caused by differential censorship between treatment arms (informative censoring). Detailed steps of the PP analyses are included in the Supplemental Methods. Second, we repeated our main analysis by restricting to patients with index dates before January 1, 2021, to ensure a minimum follow-up of 1 year. Third, we used a more stringent outcome definition of HHF by requiring hospitalization with a diagnosis code for heart failure in the first position only rather than the first or second position. Fourth, to detect the presence of potential unobserved confounding by indication, we reperformed the main analysis by using the positive control outcome of MACEs (composite of cardiovascular death, myocardial infarction, and stroke), which has a well-established nonnull causal association based on randomized clinical trials.[6,7,12] In addition, to assess the potential impact of unobserved body mass index (BMI) confounding, we conducted quantitative bias analyses using the array approach proposed by D'Agostino McGowan.[32] Specifically, we computed bias-adjusted HRs across plausible parameter ranges for (1) between-group BMI differences (GLP-1RA versus DPP-4i or SGLT-2i users) and (2) the magnitude of the association of BMI with HHF (detailed information is provided in the Supplemental Methods). Last, we repeated our analyses using the revised Lund-Malmo[33] and new European Kidney Function Consortium equations[34] to estimate GFR from creatinine. Both equations have shown better performance in Swedish populations than the Chronic Kidney Disease Epidemiology Collaboration 2021 equation.[26] Furthermore, the revised Lund-Malmo equation is used in clinical practice in Sweden.

RESULTS

Patient Characteristics

We identified 116 572 adult patients with T2D in the SCREAM database. After application of inclusion and exclusion criteria, there were 32 979 adults with T2D who newly initiated a GLP-1RA or DPP-4i in target trial 1 and 30 104 adults with T2D who initiated a GLP-1RA or SGLT-2i in target trial 2 (patient inclusion flowcharts shown in Figure S2).

In the GLP-1RA versus DPP-4i cohort, the mean age was 64 years, 40% were female, 28% had a history of atherosclerotic cardiovascular disease, and 10% had a history of heart failure (Table S4). Commonly used medications included metformin (79%), insulin (22%), angiotensin-converting enzyme inhibitor/angiotensin receptor blockers (65%), lipid-lowering therapy (61%), and β-blockers (39%). Mean HbA1c was 8.1%; median eGFR was 88 mL·min^{-1}·1.73 m^{-2}; and median UACR was 16 mg/g. Compared with those starting DPP-4i, patients starting GLP-1RA were younger (60 years versus 66 years of age), were more likely to be single/widowed/divorced (47% versus 38%), had a higher education level (secondary school, 46% versus 41%; university, 29% versus 29%), and had a higher prevalence of insulin (32%

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versus 15%) and SGLT-2i (11% versus 5%) use (Table S4). Furthermore, they had a higher mean HbA1c (8.2% versus 8.0%) and median eGFR (92 mL·min^{-1}·1.73 m^{-2} versus 85 mL·min^{-1}·1.73 m^{-2}), with no substantial differences in comorbidities. Among GLP-1RA initiators, 59% started liraglutide and 33% started semaglutide. Among DPP-4i initiators, 84% started sitagliptin and 15% started linagliptin (Table S5 and Figure S3).

Overall baseline characteristics in target trial 2 were comparable to those in target trial 1. GLP-1RA initiators were younger than those initiating an SGLT-2i (61 years versus 65 years of age), were more likely to be single/widowed/divorced (46% versus 40%), and had a lower prevalence of atherosclerotic cardiovascular disease (26% versus 36%) and heart failure (9% versus 13%) and a lower prevalence of associated cardiovascular preventative medications (Table S6). The most commonly used SGLT-2is were empagliflozin (75%) and dapagliflozin (25%) (Table S5 and Figure S3).

After inverse probability of treatment weighting, all baseline characteristics in the 2 target trials were well balanced, with standardized mean differences <0.10 (Table 2, Tables S4 and S6, and Figures S4 and S5). The distribution of propensity scores and weights is shown in Figures S6 through S9, which showed sufficient overlap of the propensity score distributions and no excessively large weights.

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Risk of HHF for GLP-1RA Versus DPP-4i and GLP-1RA Versus SGLT-2i

In target trial 1, the weighted incidence rates were 12.1 versus 15.8 events per 1000 person-years for GLP-1RA versus DPP-4i, respectively (Table 3). The weighted 3-year absolute risks of HHF were 3.4% versus 4.3%, corresponding to a weighted 3-year RD of −0.9% (95% CI, −1.5% to −0.3%) and a weighted HR of 0.77 (95% CI, 0.66–0.91; Figure 1A and Figure S10A and S10B).

In target trial 2, the weighted incidence rates were 12.8 versus 12.6 events per 1000 person-years for GLP-1RA versus SGLT-2i, respectively (Table 3). The weighted 3-year absolute risks of HHF were 3.6% versus 3.3%, corresponding to a weighted 3-year RD of 0.2% (95% CI, −0.4% to 0.8%) and a weighted HR of 1.02 (95% CI, 0.85–1.18; Figure 1B and Figure S10C and S10D).

Analyses of Single GLP-1RA Agents, Effect Modification by Baseline Heart Failure Risk, and Subgroup Analyses

Compared with DPP-4i, both liraglutide and semaglutide were associated with a lower rate of HHF, with HRs of 0.77 (95% CI, 0.66–0.93) and 0.59 (95% CI, 0.31–0.94), respectively (Table 4, Figure 2, and Figures S10 and S11). Furthermore, liraglutide and semaglutide were not associated with a lower rate of HHF compared with SGLT-2i, with HRs of 0.94 (95% CI, 0.79–1.14) and 0.80 (95% CI, 0.52–1.08), respectively (Table 4, Figure 2, and Figures S11 and S12).

When the analysis was restricted to patients without a history of heart failure, the absolute RD in HHF between GLP-1RA and DPP-4i on HHF was largest for those with a higher predicted baseline heart failure risk; for example, the 3-year RD was −0.06% (95% CI, −0.41% to 0.35%) for those with a TRS-HF$_{DM}$ score of 0 and −1.77% (95% CI, −3.23% to −0.13%) for those with a TRS-HF$_{DM}$ score ≥2 (Figure S13). In contrast, this risk-dependent trend was not observed for MACEs (Figure S14).

The association between GLP-1RA versus DPP-4i and HHF was consistent across the majority of subgroups, with no statistically significant multiplicative interactions observed (Figure S15). There was no association between GLP-1RA versus SGLT-2i and HHF among all subgroups, congruent with the main analysis (Figure S16).

Sensitivity Analyses

At 12 months of follow-up, 34% of GLP-1RA initiators and 38% DPP-4i initiators in target trial 1 and 36% of GLP-1RA initiators and 40% of SGLT-2i initiators in target trial 2 discontinued or switched their initial treatments (Table S7). The results of the PP analyses were consistent with the main analysis, although HRs and RDs were larger. The 3-year RD was −1.8% (95% CI, −2.5% to −1.1%) lower for GLP-1RA versus DPP-4i, with a weighted HR of 0.62 (95% CI, 0.50–0.75). In contrast, for GLP-1RA versus SGLT-2i, the 3-year RD (0.2% [95% CI, −0.5% to 1.0%]) and weighted HR (1.07 [95% CI, 0.85–1.34]) were comparable (Figure 1C and 1D, Table S8, and Figure S17). Results were also consistent with our main analysis when the analysis was restricted to patients with index dates before January 1, 2021, and when diagnosis codes for HHF in the first position were used (Tables S9 and S10).

In line with previous trial and observational evidence, we observed lower MACE risks for GLP-1RA versus DPP-4i (weighted HR, 0.85 [95% CI, 0.74–0.99]) and similar MACE risks for GLP-1RA versus SGLT-2i (weighted HR, 0.98 [95% CI, 0.85–1.13]; Table 3). Quantitative bias analysis showed that bias-adjusted HRs exhibited minimal deviation from our main findings across 18 parameter combinations of (1) BMI differences between GLP-1RA and DPP-4i or SGLT-2i users and (2) strength of the BMI-HHF association (Table S11). The results obtained with different eGFR equations were consistent with the main analysis (Table S12).

DISCUSSION

Statement of Principal Findings

In this large cohort of patients with T2D treated by routine care in Sweden, we found that GLP-1RA use was

Table 2. Selected Baseline Characteristics for GLP-1RA* Versus DPP-4i* (Target Trial 1) and GLP-1RA* Versus SGLT-2i* (Target Trial 2) After Inverse Probability of Treatment Weighting

	Target trial 1: GLP-1RA vs DPP-4i		Target trial 2: GLP-1RA vs SGLT-2i	
	GLP-1RA	DPP-4i	GLP-1RA	SGLT-2i
n	13 803	19 164	14 774	15 347
Demographic characteristic				
Age, mean (SD), y	63 (12)	64 (13)	63 (12)	63 (12)
Female, n (%)	5512 (39.9)	7737 (40.4)	5577 (37.7)	5799 (37.8)
Duration of diabetes, median (IQR), y	6.6 (2.5–11.2)	6.7 (2.6–11.2)	7.5 (3.1–12.5)	7.6 (3.0–12.5)
Laboratory measurements				
HbA1c, mean (SD), %	8.1 (1.6)	8.1 (1.6)	8.2 (1.6)	8.2 (1.6)
eGFR, median (IQR), mL·min^{-1}·1.73 m^{-2}	88 (70–99)	87 (68–99)	89 (73–100)	89 (73–100)
eGFR <60 mL·min^{-1}·1.73 m^{-2}, n (%)	2320 (16.7)	3367 (17.5)	1888 (12.8)	1955 (12.8)
UACR, median (IQR), mg/g	15 (6–34)	17 (7–34)	14 (6–32)	14 (6–32)
Albuminuria category, n (%)				
A1	10 135 (73.4)	13 992 (73.0)	10 913 (73.9)	11 337 (73.9)
A2	2763 (20.0)	3859 (20.1)	2982 (20.2)	3087 (20.1)
A3	906 (6.6)	1314 (6.9)	879 (5.9)	923 (6.0)
Medical history, n (%)				
Hypertension	9901 (71.7)	13 858 (72.3)	11 168 (75.6)	11 597 (75.6)
Atherosclerotic cardiovascular disease	3800 (27.5)	5375 (28.0)	4555 (30.8)	4760 (31.0)
Heart failure	1312 (9.5)	1870 (9.8)	1598 (10.8)	1648 (10.7)
Atrial fibrillation	1455 (10.5)	2060 (10.8)	1673 (11.3)	1753 (11.4)
Medication use, n (%)				
ACEi/ARB	8959 (64.9)	12 443 (64.9)	9924 (67.2)	10 348 (67.4)
MRA	728 (5.3)	1033 (5.4)	952 (6.4)	976 (6.4)
β-Blocker	5281 (38.3)	7417 (38.7)	5939 (40.2)	6219 (40.5)
Thiazide diuretic	3400 (24.6)	4791 (25.0)	3667 (24.8)	3815 (24.9)
Antiplatelet therapy	3739 (27.1)	5229 (27.3)	4096 (27.7)	4251 (27.7)
Lipid-lowering drug	8349 (60.5)	11 627 (60.7)	9610 (65.0)	9955 (64.9)
Diabetes drugs in previous 6 mo, n (%)				
Metformin	10 989 (79.6)	15 292 (79.8)	11 785 (79.8)	12 238 (79.7)
Sulfonylurea	3021 (21.9)	4161 (21.7)	2742 (18.6)	2857 (18.6)
SGLT-2i	1128 (8.2)	1635 (8.5)	…	…
DPP-4i	…	…	3090 (20.9)	3140 (20.5)
Insulin	3136 (22.7)	4481 (23.4)	3915 (26.5)	4112 (26.8)

ACEi indicates angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; DPP-4i, dipeptidyl peptidase-4 inhibitor; eGFR, estimated glomerular filtration rate; GLP-1RA, glucagon-like peptide-1 receptor agonist; HbA1c, glycated hemoglobin; IQR, interquartile range; MRA, mineralocorticoid receptor antagonist; SGLT-2i, sodium-glucose cotransporter-2 inhibitor; and UACR, urinary albumin-to-creatinine ratio.

*GLP-1RAs included exenatide, liraglutide, lixisenatide, dulaglutide, and semaglutide. DPP-4i included sitagliptin, vildagliptin, and linagliptin, with saxagliptin excluded because of prior safety concerns regarding heart failure. SGLT-2i included dapagliflozin, canagliflozin, empagliflozin, and ertugliflozin. Other specific agents within these classes (eg, albiglutide, sotagliflozin, alogliptin) were not available in Sweden during the study period.

associated with a lower rate of HHF compared with DPP-4i use. Furthermore, we found that rates for HHF were similar when comparing GLP-1RA initiators and new users of SGLT-2i. Our findings were consistent across most subgroups, as well as several sensitivity analyses, and among comparisons of single agents within the GLP-1RA class (ie, semaglutide and liraglutide).

Novelty and Comparison With Previous Studies

Our study provides several new findings and builds on current evidence. First, there is conflicting evidence on the effects of GLP-1RA on HHF in patients with T2D. A meta-analysis including the 8 large cardiovascular outcome trials showed a small relative rate reduction

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Table 3. Number of Events, Weighted Incidence Rates, Weighted Absolute Risks, and Weighted HRs for HHF and MACEs Among New Users of GLP-1RA* Versus DPP-4i* and Among New Users of GLP-1RA*Versus SGLT-2i* in Intention-to-Treat Analyses

Treatment arm	No. of individuals	No. of events	Median follow-up (IQR), mo	Weighted incidence rate per 1000 PY (95% CI)	3-y absolute risk (95% CI), %	3-y risk difference (95% CI), %	Crude HR (95% CI)	Weighted HR (95% CI)
Primary outcome: HHF								
Target trial 1: GLP-1RA vs DPP-4i								
DPP-4i	18 972	801	36.0 (24.0 to 36.0)	15.8 (14.6 to 17.0)	4.30 (3.98 to 4.67)	Reference	Reference	Reference
GLP-1RA	14 007	310	26.0 (9.9 to 36.0)	12.1 (10.9 to 13.4)	3.43 (2.96 to 3.95)	−0.88 (−1.45 to −0.25)	0.64 (0.56 to 0.73)	0.77 (0.66 to 0.91)
Target trial 2: GLP-1RA vs SGLT-2i								
SGLT-2i	15 229	390	21.0 (8.2 to 36.0)	12.6 (11.3 to 13.9)	3.33 (2.94 to 3.70)	Reference	Reference	Reference
GLP-1RA	14 875	342	26.0 (10.9 to 36.0)	12.8 (11.5 to 14.1)	3.56 (3.14 to 3.97)	0.23 (−0.35 to 0.76)	0.82 (0.71 to 0.94)	1.02 (0.85 to 1.18)
Positive control outcome: MACEs†								
Target trial 1: GLP-1RA vs DPP-4i								
DPP-4i	18 958	1144	36.0 (23.7 to 36.0)	22.6 (21.2 to 24.1)	6.31 (5.92 to 6.72)	Reference	Reference	Reference
GLP-1RA	13 999	429	26.0 (9.9 to 36.0)	19.2 (17.7 to 20.8)	5.49 (4.85 to 6.19)	−0.82 (−1.59 to 0.08)	0.64 (0.57 to 0.71)	0.85 (0.74 to 0.99)
Target trial 2: GLP-1RA vs SGLT-2i								
SGLT-2i	15 202	514	21.0 (8.1 to 36.0)	17.5 (16.0 to 19.1)	4.91 (4.44 to 5.46)	Reference	Reference	Reference
GLP-1RA	14 866	441	26.0 (10.9 to 36.0)	17.2 (15.7 to 18.8)	4.88 (4.39 to 5.39)	−0.03 (−0.72 to 0.66)	0.79 (0.69 to 0.89)	0.98 (0.85 to 1.13)

DPP-4i indicates dipeptidyl peptidase-4 inhibitor; GLP-1RA, glucagon-like peptide-1 receptor agonist; HHF, hospitalization for heart failure; HR, hazard ratio; IQR, interquartile range; MACE, major adverse cardiovascular event; PY, person-years; and SGLT-2i, sodium-glucose cotransporter-2 inhibitor.

*GLP-1RA included exenatide, liraglutide, lixisenatide, dulaglutide, and semaglutide. DPP-4i included sitagliptin, vildagliptin, and linagliptin, with saxagliptin excluded because of prior safety concerns regarding heart failure. SGLT-2i included dapagliflozin, canagliflozin, empagliflozin, and ertugliflozin. Other specific agents within these classes (eg, albiglutide, sotagliflozin, alogliptin) were not available in Sweden during the study period.

†Per our exclusion criterion for patients with <1 day of follow-up, the MACE analysis had more exclusions for events on the index date than the HHF analysis. Specifically, MACEs/death led to 62 (target trial 1, 61 MACEs, 1 noncardiovascular death) and 142 (target trial 2, 142 MACEs) exclusions versus 40 (target trial 1, 39 HHF, 1 death) and 106 (target trial 2, 105 HHF, 1 death) for HHF.

(HR, 0.89 [95% CI 0.82–0.98]),[6] whereas another meta-analysis that also included smaller trials showed no association (HR, 0.94 [95% CI, 0.85–1.03]),[12] with both estimates being close to the null. Nevertheless, recent trials in different populations have shown strong reductions in HHF for GLP-1RA compared with placebo, including among those with obesity,[9] diabetic kidney disease,[10] and heart failure with preserved ejection fraction,[11] as well as recent observational studies.[35] Therefore, a recent meta-analysis that included these recent trials observed an incidence rate ratio of 0.85 (95% CI, 0.75–0.97) for HHF.[36] Our findings indicate that GLP-1RAs are also associated with a lower risk of HHF in patients with T2D compared with DPP-4is. This is congruent with a recent meta-analysis that found that GLP-1RAs reduced HHF only among those without baseline heart failure, not in those with heart failure.[37] Indeed, in our population, ≈90% of patients did not have heart failure. A second novelty is that our observational design allowed us to study the head-to-head comparative effectiveness of GLP-1RA versus SGLT-2i, whereas randomized trials have focused on comparisons with placebo. We found that GLP-1RAs and SGLT-2is were associated with similar rates of HHF. Although robust randomized evidence supports the benefit of SGLT-2i for preventing HHF, comparable evidence for GLP-1RAs is less well established but emerging. A recent trial suggests strong risk reductions in HHF for GLP-1RAs (with or without glucose-dependent insulinotropic polypeptide) in heart failure with preserved ejection fraction or diabetic kidney disease.[10,11,38] Third, our large sample size allowed us to study single GLP-1RA agents, and we found that both liraglutide and semaglutide, the most commonly used GLP-1RA agents, were associated with a lower risk of HHF versus DPP-4i. Notably, in the semaglutide analyses, the cumulative incidence curves separated early, raising the possibility of a conditional exchangeability violation between treatment groups, although this pattern may also reflect chance. Our estimated HR of 0.59 (95% CI, 0.31–0.94) is broadly consistent with prior randomized trial evidence, but direct comparisons

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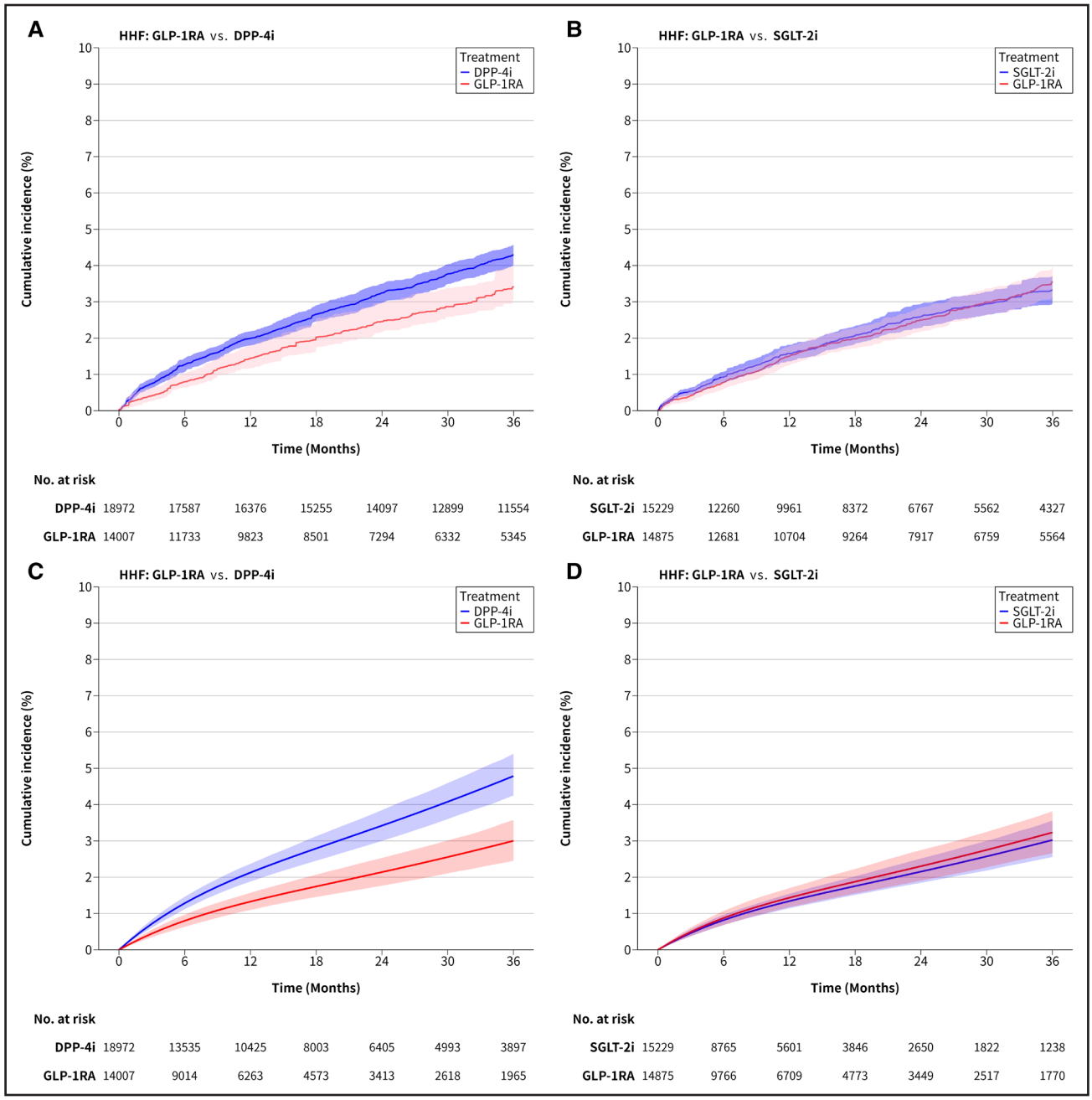

Figure 1. Weighted cumulative incidence curves for GLP-1RA* versus DPP-4i* and GLP-1RA* versus SGLT-2i* on HHF in (A and B) intention-to-treat and (C and D) per-protocol analyses.

DPP-4i indicates dipeptidyl peptidase-4 inhibitor; GLP-1RA, glucagon-like peptide-1 receptor agonist; HHF, hospitalization for heart failure; and SGLT-2i, sodium-glucose cotransporter-2 inhibitor. *GLP-1RA included exenatide, liraglutide, lixisenatide, dulaglutide, and semaglutide. DPP-4i included sitagliptin, vildagliptin, and linagliptin, with saxagliptin excluded because of prior safety concerns regarding heart failure. SGLT-2i included dapagliflozin, canagliflozin, empagliflozin, and ertugliflozin. Other specific agents within these classes (eg, albiglutide, sotagliflozin, alogliptin) were not available in Sweden during the study period.

are difficult because of differences in patient populations and the specific GLP-1RA studied. For example, the most recent meta-analysis of GLP-1RA trials has reported incidence rate ratios of 0.85 (95% CI, 0.75–0.97) for HHF overall,[36] and a recent semaglutide-focused meta-analysis in patients with heart failure with mildly reduced ejection fraction/heart failure with preserved ejection fraction yielded an HR of 0.59 (95% CI, 0.41–0.82).[39] Fourth, our observational study extends trial evidence by including a diverse population that reflects routine clinical care. Although the large cardiovascular outcome trials included patients with cardiovascular disease or at high cardiovascular risk, patients in routine clinical practice are more heterogeneous, with differing levels of baseline cardiovascular risk and comorbid conditions. In our study, the baseline

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Table 4. Number of Events, Incidence Rates, Absolute Risks, and HRs for HHF Events Among New Users of liraglutide Versus DPP-4i,* Liraglutide Versus SGLT-2i,* Semaglutide Versus DPP-4i,* and Semaglutide Versus SGLT-2i* in Intention-to-Treat Analyses

Treatment arm	No. of individuals	No. of events	Median follow-up (IQR), mo	Weighted incidence rate per 1000 PY (95% CI)	2 or 3-y absolute risk (95% CI),† %	2 or 3-y risk difference (95% CI),†%	Crude HR (95% CI)	Weighted HR (95% CI)
Liraglutide vs DPP-4i								
DPP-4i	18972	801	36.0 (24.0 to 36.0)	16.0 (14.9 to 17.1)	4.42 (4.08 to 4.75)	Reference	Reference	Reference
Liraglutide	8220	235	36.0 (25.0 to 36.0)	12.3 (10.9 to 13.9)	3.49 (2.96 to 4.11)	−0.92 (−1.49 to −0.22)	0.67 (0.58 to 0.77)	0.77 (0.66 to 0.93)
Liraglutide vs SGLT-2i								
SGLT-2i	15229	390	21.0 (8.2 to 36.0)	12.9 (11.7 to 14.2)	3.46 (3.02 to 3.84)	Reference	Reference	Reference
Liraglutide	8829	252	36.0 (25.0 to 36.0)	12.1 (10.6 to 13.8)	3.43 (2.95 to 3.93)	−0.04 (−0.63 to 0.56)	0.83 (0.71 to 0.97)	0.94 (0.79 to 1.14)
Semaglutide vs DPP-4i								
DPP-4i	8272	293	24.0 (12.0 to 24.0)	21.5 (18.9 to 24.4)	3.82 (3.32 to 4.31)	Reference	Reference	Reference
Semaglutide	4685	43	9.0 (3.0 to 14.0)	12.8 (10.0 to 16.3)	2.61 (1.15 to 4.73)	−1.21 (−2.83 to 0.83)	0.41 (0.30 to 0.57)	0.59 (0.31 to 0.94)
Semaglutide vs SGLT-2i								
SGLT-2i	12504	281	15.9 (7.0 to 24.0)	17.0 (15.0 to 19.3)	3.01 (2.64 to 3.39)	Reference	Reference	Reference
Semaglutide	4812	51	9.0 (4.0 to 14.9)	14.0 (10.9 to 17.6)	2.41 (1.47 to 3.45)	−0.60 (−1.57 to 0.59)	0.63 (0.46 to 0.85)	0.80 (0.52 to 1.08)

DPP-4i indicates dipeptidyl peptidase-4 inhibitor; GLP-1RA, glucagon-like peptide-1 receptor agonist; HHF, hospitalization for heart failure; HR, hazard ratio; IQR, interquartile range; PY, person-years; and SGLT-2i, sodium-glucose cotransporter-2 inhibitor.

*DPP-4i included sitagliptin, vildagliptin, and linagliptin, with saxagliptin excluded because of prior safety concerns regarding heart failure. SGLT-2i included dapagliflozin, canagliflozin, empagliflozin, and ertugliflozin. Other specific agents within these classes (eg, sotagliflozin, alogliptin) were not available in Sweden during the study period.

†For semaglutide versus DPP-4i or SGLT-2i, follow-up was truncated at 2 years; therefore, we report 2-year (not 3-year) absolute risk and risk difference.

prevalence of atherosclerotic cardiovascular disease was 28%, and it ranged between 31% and 100% in the cardiovascular outcome trials. Indeed, the baseline rate of MACEs (≈20 per 1000 person-years) in our cohort was lower than that of all cardiovascular outcome trials in T2D, which ranged from 24 to 64 events per 1000 person-years.[40–46] The incidence rate of HHF in our study (≈12 per 1000 person-years) was similar to trials such as LEADER (Liraglutide Effect and Action in Diabetes: Evaluation of Cardiovascular Outcome Results; ≈12 per 1000 person-years) and PIONEER 6 (Peptide Innovation for Early Diabetes Treatment 6; ≈10 per 1000 person-years) but substantially lower than FLOW (≈38 per 1000 person-years), which is likely explained by the requirement for chronic kidney disease in the FLOW trial (Table S13). We showed that the association between GLP-1RA use and primary prevention of HHF was strongest among patients with the largest predicted risk of developing heart failure as assessed by the TRS-HF_{DM} score. It is important to note that this observation aligns with the paradigm of using risk tools to identify individuals who are most likely to benefit from primary preventive therapies, as recently articulated by the American Heart Association.[30]

Possible Explanations and Clinical Implications

Our findings suggest that GLP-1RAs could be used to reduce the risk of HHF in patients with T2D, regardless of heart failure history, thereby addressing both primary and secondary prevention needs. These benefits are in addition to their established protection against MACEs. This is especially important because T2D is a strong risk factor for developing heart failure, and the number of patients with T2D is expected to increase as a result of population aging and a rise in risk factors such as obesity and comorbidities such as diabetic kidney disease. The mechanisms by which GLP-1RAs exert their cardiovascular beneficial effects may include those related to weight loss, which leads to improved glucose levels and improvements in cardiovascular risk factors such as blood pressure, inflammation, and lipid levels and other direct effects on the cardiovascular system and kidney function.[47,48]

Unanswered Questions and Future Research

Our main findings need to be replicated in future high-quality studies. Our study included few patients with heart failure at baseline, and recent evidence suggests that there may be differential effects of GLP-1RAs among

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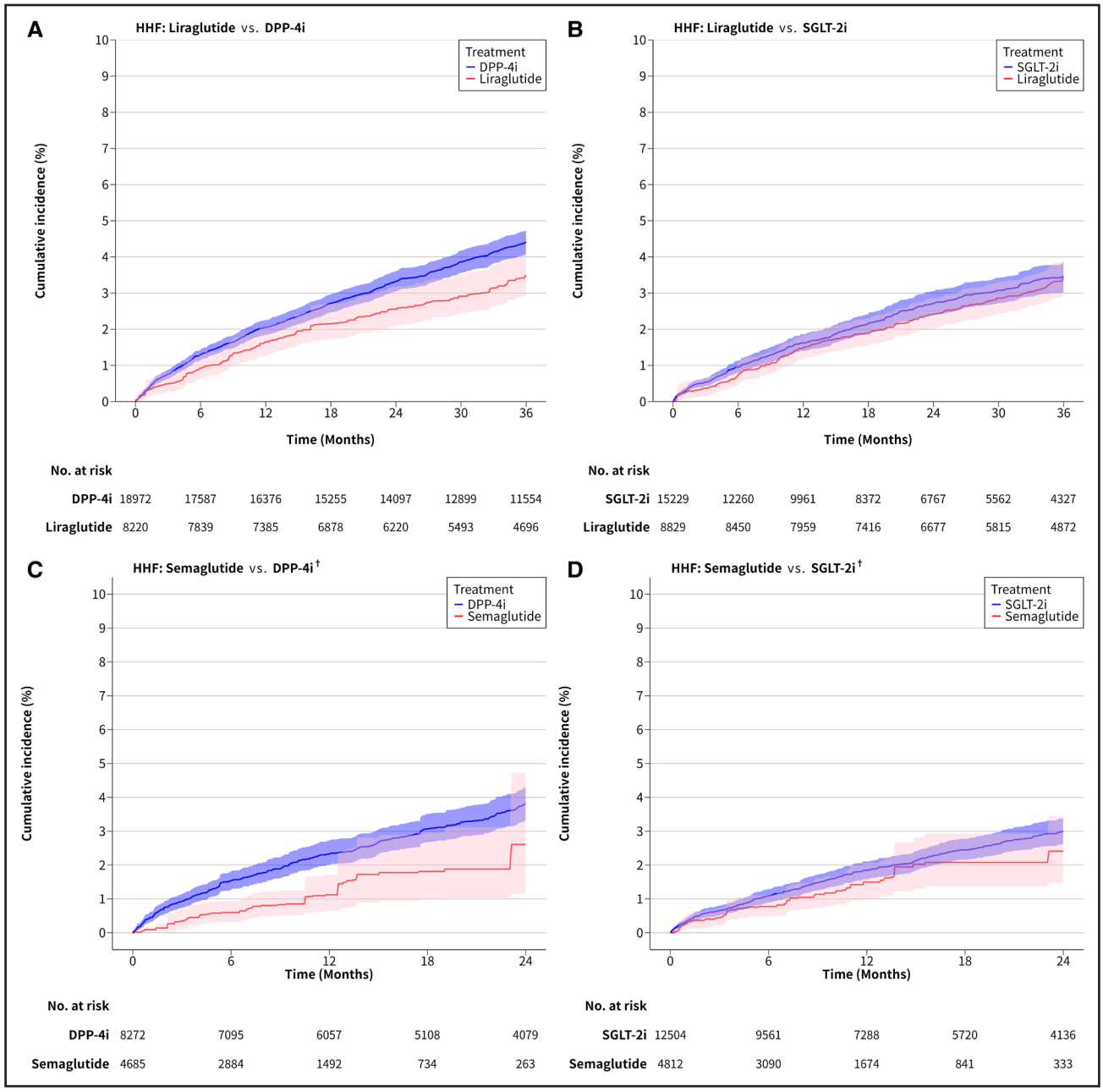

Figure 2. Weighted cumulative incidence curves for (A) liraglutide versus DPP-4i,* (B) liraglutide versus SGLT-2i,* (C) semaglutide versus DPP-4i,* and (D) semaglutide versus SGLT-2i* on HHF in intention-to-treat analyses.
DPP-4i indicates dipeptidyl peptidase-4 inhibitor; HHF, hospitalization for heart failure; and SGLT-2i, sodium-glucose cotransporter-2 inhibitor. *DPP-4i included sitagliptin, vildagliptin, and linagliptin, with saxagliptin excluded because of prior safety concerns regarding heart failure. SGLT-2i included dapagliflozin, canagliflozin, empagliflozin, and ertugliflozin. Other specific agents within these classes (eg, sotagliflozin, alogliptin) were not available in Sweden during the study period. †For semaglutide versus DPP-4i or SGLT-2i, follow-up was truncated at 2 years.

those with diabetes and heart failure with preserved ejection fraction or heart failure with reduced ejection fraction.

Strengths and Weaknesses of the Study

Strengths of our study include its large sample size and application of the target trial emulation framework to structure the application of causal inference methods. Furthermore, we adopted an active comparator new user design to reduce confounding and to prevent time-related and selection bias caused by prevalent users.[20]

Our study also has limitations. First, we cannot rule out the presence of unmeasured confounding in our study, although we implemented a number of steps to mitigate it as much as possible. We adjusted for a wide range of confounders, including laboratory measurements such as HbA1c, eGFR, and UACR and socioeconomic status variables. Furthermore, our positive control analysis was

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congruent with findings from randomized controlled trials, with our HR of MACEs of 0.85 being close to that of a recent meta-analysis[6] (HR, 0.86), indicating no major residual confounding. Although BMI data were unavailable in our analyses, GLP-1RA users are likely to have had higher baseline BMI than the comparator groups, given the known weight-loss effects of GLP-1RAs. This would likely bias effect estimates away from the null rather than attenuate them. Second, we had missing data in some laboratory measurements such as eGFR and UACR (ranging between 10.3% and 31.5%). We therefore used multiple imputation under the assumption that the missingness could be accurately predicted by the other covariates in our imputation model. Third, we could not distinguish heart failure with preserved ejection fraction and heart failure with reduced ejection fraction because of missing left ventricular ejection fraction data in the SCREAM database, which limited our ability to perform subtype-specific analyses. Fourth, discontinuation rates of GLP-1RA, DPP-4i, and SGLT-2i were high, which is consistent with other studies from routine clinical practice.[49,50] When a large proportion of patients discontinue therapy, intention-to-treat effects may be biased toward the null. To address this, we also conducted a PP analysis, which estimates the effect of remaining on treatment throughout follow-up. Fifth, our findings are representative of the population in Stockholm but may not be generalizable to other settings. For example, geographic variation in the incidence of HHF can influence the absolute benefit of GLP-1RAs because lower baseline risk leads to smaller absolute risk reductions. Sixth, we lacked race and ethnicity data because collecting such data is prohibited in Sweden by law. Last, we evaluated a risk-based approach for the primary prevention of heart failure using the TRS-HF_{DM} risk score. Although this model has been specifically developed and validated for patients with T2D, the PREVENT-HF (Predicting Risk of Cardiovascular Disease EVENTs-Heart Failure) models are generally recommended for implementation in the United States.[30] However, because of a lack of data on key variables such as smoking status, BMI, and blood pressure, we were unable to calculate the PREVENT-HF risk score.

Conclusions

We found that GLP-1RAs were associated with a lower rate of HHF compared with DPP-4is and similar rates compared with SGLT-2is in a broad unselected population of patients with T2D, including those with and without a history of heart failure. These findings suggest that GLP-1RAs could be used for the primary and secondary prevention of heart failure.

ARTICLE INFORMATION

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Author Contributions

Y.X. had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis. Concept and design: Y.X., J.-J.C., and E.L.F. Acquisition of data: J.-J.C. Analysis and interpretation of data: all authors. Drafting of the manuscript: Y.X. and E.L.F. Critical revision of the manuscript for important intellectual content: all authors. Statistical analysis: Y.X., T.H., Y.Z., D.J., and E.L.F. Administrative, technical or material support: all authors. Supervision: E.L.F. and Y.X.

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Disclosures

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Supplemental Material

Checklist
Expanded Methods
Tables S1–S13
Figures S1–S17
References 51–61

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